

Glass Is an Unclosed Layer: Kovacs Memory as Predictive-Quotient Residue

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Abstract

A glass and its equilibrium liquid can agree in every routinely monitored quantity and still evolve differently. Kovacs’s 1963 crossover experiment makes the point vivid: at the moment an aged sample’s volume equals its equilibrium value, the volume moves away again. We study the question behind this on finite Markov models small enough to settle it by exact computation (the East and Fredrickson–Andersen kinetically constrained spin chains and a Bouchaud-type trap family): is a declared readout, here the energy, a *closed* description, one whose present value fixes its own future statistics? On the declared East scopes where we certify it, it is not, and we certify this in several independent ways. For an aging East chain, the energy readout’s closure deficit is at least $3/2$ times its constrained-equilibrium value at every tested horizon, certified in exact rational arithmetic, and the equilibrium value is itself provably nonzero. The Kovacs protocol yields two preparations with the same mean energy now and different mean energies later. Two randomized stops of one aging run share the *entire* energy distribution, exactly, yet their mean energies separate at each of 21 later horizons, so no function of the current energy law, a fictive-temperature-like index included, predicts the future on that shell. All 512 yes/no functions of the energy agree on a further, constructed pair that one structural coordinate separates. A kill-test panel rejects five look-alike regimes, and one added coordinate resolves each exhibited pair. Aging constants frozen on East and scored mechanically on FA pass 8 of 22 predictions; the trap column is empty for a reason we prove. Lean checks the abstract theorems and the exported witness tables. We also report what does not hold: the fixed-point route to a full strict-extension filing fails here, and a positive infinite-time conditional pressure gap is provably zero. Every claim is pinned by content hash to its evidence, and every nonclaim is printed.

Keywords. glass transition; Kovacs effect; aging and memory; kinetically constrained models; East model; fictive temperature; closure deficit; lumpability; predictive quotient; Six Birds Theory; emergence calculus; exact finite certificates; pre-registered prediction; Lean.

1 Introduction

1.1 The problem

Anderson called the nature of glass and the glass transition probably the deepest and most interesting unsolved problem in solid-state theory [3]. One face of that problem is sixty years old and unusually concrete: no universally accepted static order parameter for glass has emerged. A glass and a liquid can agree in density, energy, and every routinely monitored one-time observable, yet behave differently [26]. Conventional low-order static correlations show no clear signature of the transition, and the candidate structural measures that do grow remain contested [8, 19, 68, 61].

A second face has sat in the laboratory record since 1963: Kovacs’s crossover experiment [40]. Equilibrate a polymer glass hot, quench it, age it, then jump the temperature up and watch its volume relax. At some moment the volume *equals* its new equilibrium value, and it does not stay there: it overshoots through a non-monotone hump before returning. At the crossing moment the sample is indistinguishable from equilibrium in the monitored observable and distinguishable from it in its future, a point made explicit at the level of configurations in simulation [46]. The standard partial fix, the Tool–Narayanaswamy–Moynihan (TNM) fictive temperature [69, 49, 47], adds one memory variable. Richer protocols expose the limits of any single such variable and have motivated multiparameter models [58, 41, 33, 43]. Memory effects of this kind, in which a material’s response depends on its history in a way that no monitored one-time observable records, are now studied across many classes of matter [37].

1.2 The question, made precise

Both faces ask the same question. Take a declared readout of the system (here, the energy of a spin chain; in the laboratory, the volume) and ask: *is that readout a closed description of the dynamics?* A readout is closed over a horizon τ if its present value determines the statistics of its value τ steps later. Then a law can be written in the readout’s vocabulary without loss; this is the informational closure of [10, 54]. In Markov-chain language the readout is closed at horizon τ exactly when the τ -step readout transition probabilities agree across every populated microstate with the same readout. For a fully supported distribution this is lumpability of the readout [17, 38]. The Kovacs hump is a direct demonstration that the monitored volume is not closed.

We use “static order parameter” in one operational sense only: a function of the declared readout that separates two preparations whose futures differ. A function of the readout that fails to separate them (a constant does so trivially) cannot carry the information the futures carry. That is narrower than the field’s usage, and it is the only sense in which this paper speaks of foreclosing the search.

Put this way, three further questions become computable on a finite model. How far from closed is the readout, in nats? Can one exhibit a concrete certificate of the failure, two preparation histories that agree on the readout now and differ in their futures? And can *any* function of the readout recover the missing distinction, or must a genuinely new coordinate carry it, and at what cost?

1.3 What this paper does

We answer these questions on small finite Markov models. The kinetically constrained East model [35, 59, 31] is the substrate on which everything is built and calibrated; the kinetically constrained Fredrickson–Andersen model [29] and a Bouchaud-type trap family [12, 44] are out-of-sample transfer targets. At the sizes used ($N \leq 10$ spins for certificates) the transition kernels, stationary measures, witness pairs and zero controls are ratios of integers. Where a certificate needs logarithms, as the closure deficit does, every inequality is proved with rational interval enclosures. The price is that every result is a statement about a declared finite class of models, sizes, observables and protocols, and we state that class in every claim.

The bookkeeping that keeps those scopes honest comes from a certificate-based closure calculus, Six Birds Theory (SBT) [77, 70, 71, 72, 73, 74, 75, 76]. The reader does not need that corpus. §2 defines every object in ordinary probabilistic terms, and the framework’s labels appear only where a claim is filed in its vocabulary, always with a plain gloss.

Table 1 lists the results. Each is stated precisely, at its declared scope, in §4–§10; the bracketed labels there, such as [GT1], are machine-checked citations into the claims ledger (§2).

In more detail:

Table 1. Results at a glance. “Theorem” means proved or exhaustively audited on the stated finite class; “certificate” means valid on a frozen shell of eligibility conditions with no broader transfer; “run-only” means a measured readout with no model fitted to it.

Result	What is shown	Grade
GT1	The aging East energy readout is far from closed: its closure deficit is at least $3/2$ times the constrained-equilibrium value at every tested horizon, and that equilibrium value is itself nonzero.	theorem
GT2	Kovacs witness: two preparations with the same mean energy now, different mean energies five steps later.	theorem
GT3	All 512 yes/no functions of the energy agree on a pair that one structural coordinate separates.	theorem
GT4	A deterministic classifier accepts the Kovacs candidate and rejects five constructed look-alike regimes.	theorem
GT5	One added coordinate resolves the Kovacs pair; three declared scalars each do.	theorem
GT6	Two preparations share the entire current energy distribution yet separate at 21 future horizons: the predictive description does not factor through the energy law. The fuller fixed-point filing does not pass.	certificate
GT7	Aging constants frozen on East and scored on FA: 8 pass, 14 fail; the trap column is structurally empty.	run-only
Lean	Abstract theorems and the exported witness tables are machine-checked; a legacy formal core provably admits no witness at all.	verified / finding

- **GT1, the energy readout is not closed.** The closure deficit is the conditional mutual information between the microstate now and the energy later, given the energy now. For an aging East chain it exceeds the same quantity at constrained equilibrium by a factor of at least $3/2$ at every declared horizon. The equilibrium value is not zero: closure depends on where the mobile excitations sit, not on whether the chain is stationary.
- **GT2, the Kovacs witness.** The Kovacs protocol on East yields an exact pair of preparation histories with the same present mean energy and different future mean energies under the declared continuation.
- **GT3, the order-parameter search is foreclosed on the declared class.** Each of the 512 yes/no functions of the energy agrees on a constructed witness pair (indeed the two histories have the same complete energy distribution) that a finer readout of the same microstate, the occupation of the site next to the wall, separates. Whatever separates the pair must carry information the energy readout does not.
- **GT4, the kill-test panel.** The classifier that accepts the Kovacs memory candidate rejects five schematic look-alikes: a protocol artifact, dissipation, an explicitly latent variable, an equilibrium control, and an incomplete continuation catalog.
- **GT5, the cost of repair.** One added scalar coordinate resolves the GT2 pair. A fictive-temperature-like index (the index of the grid point whose equilibrium mean energy is closest to each microstate’s energy, averaged over the history) is one of three that do. This shows the repair mechanism; it does not prove that one scalar suffices, or fails, in general.
- **GT6, the memory layer is strictly richer than the full energy law.** GT3’s pair is constructed and separated by a present readout. GT6 asks the same of real aging: on a frozen aging shell, two lawful randomized stops of one run have *identical* current energy distributions,

exactly, and the East dynamics drives their mean energies apart at every horizon from 1 to 21. So the future-equivalence description of histories is not a function of the current energy law, and no descriptor computed from that law, the fictive-temperature-like index included, is future-sufficient there. The gaps are tiny (about 10^{-10}) but exact. The framework’s fuller strict-extension filing, which also asks for multiple stable fixed points of a packaging map, does not pass on this shell, and we show why.

- **GT7, aging constants transfer only partly.** Six aging readouts were frozen on East and interval predictions for FA and the trap family were registered and hash-pinned. The targets were run and scored mechanically: 8 of 22 scored predictions passed, all 14 failures lie on the same side, and the trap family produced no Kovacs hump at all, for a reason we prove.

Two further results come from the formalization (§9). In the Lean proof assistant the abstract theorems are machine-checked, together with the exact witness tables of GT2 and GT6 as exported from the computation. And the legacy formal core we started from provably cannot express any witness at all; the theorem locates exactly which axiom causes the collapse.

1.4 How to read the claims

We would defend four contributions, in this order. First, the **discipline**: every claim quantifies over a declared finite class, is pinned by content hash to the artifact that supports it, comes with controls, and is filed together with what it does *not* assert. Honest negatives are results. Second, the **result complex** itself: the claims, bridges and honest negatives of Table 1, each hash-pinned to its artifacts. Third, the **freeze-and-transfer protocol with its actual scoreboard**: the failures are results, with a clean one-sided pattern and a proven structural explanation for the empty column. Fourth, the **Lean-checked core and its scope-boundary theorem**.

What this paper is not: it proves nothing about all glasses, the continuum, or the ideal-glass transition; it makes no irreversibility claims; and its field-facing sentence, “no static order parameter,” is licensed only over the declared class on which it is proved. The full list of nonclaims is printed, unabridged, in Appendix B.

2 Definitions and Methods

2.1 The objects, in plain terms

Everything in this paper is built from a finite-state Markov chain and a choice of what to observe. Five objects carry all the results. We define each in ordinary terms, with the Kovacs experiment as the running example, and give its source in the Six Birds Theory (SBT) corpus, which we cite rather than re-derive [77, 70, 71, 72, 73, 74, 75, 76].

Lens. What is observed: a map f from the microstate space Z to a finite set of readout values. Our main lens is the energy readout L_{energy} , the number of up spins; the “thermodynamic lens” Π_{std} is this readout together with its declared panel. In the laboratory the lens is the dilatometer. (SBT: a finite theory package, F-I [70], Def. 1.)

Closure and the closure deficit. A lens is *closed* at horizon τ , for a given law and a given distribution over microstates, if the readout now determines the distribution of the readout τ steps later, with nothing further gained from the microstate. The *closure deficit*

$$\text{CD}_\tau(\Pi) = I(X_t; Y_{t+\tau} | Y_t)$$

is the conditional mutual information, in nats [23], between the microstate X_t and the future readout $Y_{t+\tau}$ given the present readout Y_t . It is zero exactly when the τ -step readout transition probabilities are the same from every *positive-mass* microstate with the same readout. When the distribution has full support this is τ -lumpability of the (kernel, lens) pair [17, 38, 15]. With partial support, a zero deficit tests conditional independence only on the populated microstates at the declared time and horizon; zero-mass regions are never audited. A point-mass start on East $N = 2$ has $CD = 0$ although the kernel is not lumpable, a regression-tested control. This does not by itself establish weak lumpability of the aggregated process, which is a statement about the whole readout process under an initial law [62]. Lumpability at one horizon does not imply it at others, so the ladder of horizons is declared. A positive deficit is an exact lower bound on what any autonomous law at that horizon, written in the lens’s vocabulary, must discard. A readout that is not closed obeys a law with a memory kernel, in the projection-operator sense of [81, 45]. (SBT: F-III [72] T8; [75] Def. 3.1.)

Packaging map and idempotence defect. A second, coarser diagnostic. Evolve the chain τ steps, push the distribution through the lens, and lift it back to a microstate distribution by a declared rule (the “prototype” for each readout value): $E_{\tau,f}(\mu) = U_f(Q_f(\mu P^\tau))$. The *idempotence defect* $\delta_{\tau,f}$ is the largest total-variation distance, over starting microstates, between applying this map twice and applying it once. It is a saturation diagnostic and nothing more. A lumpable lens can still have $\delta_{\tau,f} > 0$: the unconstrained control at $N = 2$, $\epsilon = 1/2$, $\tau = 1$ has exact defect $2/9$, a regression test. The defect is also distinct from the *prototype-retention error*, the largest probability, over readout values, that the chain started from a value’s prototype shows a different readout τ steps later; the retention error bounds the defect. GT1 reports the defect only as a ratio between two kernels, never as a closure criterion. (SBT: F-I Def. 9.)

Preparation histories, quotients, and witnesses. A *preparation history* is a protocol word (“equilibrate at ϵ_1 , quench to ϵ_2 , age t_w , jump to ϵ_3 ”) together with the state distribution it produces. Over a declared finite catalog of histories and continuation protocols, the *current quotient* Q identifies histories with equal present lens readouts. The *predictive quotient* M identifies histories with equal future readout statistics under every declared continuation: the causal-state construction of computational mechanics [24, 63, 64], restricted to a declared finite catalog. When future equivalence implies current equivalence, M refines Q and there is a comparison map $\pi : M \rightarrow Q$. This refinement is a premise, not automatic: the Python pipeline verifies it on every declared carrier before forming π , and in the Lean cores it follows because the identity continuation and the present events are among the declared futures. A *witness*, or split pair, is two histories h, h' that Q identifies and M separates: same now, different later (Fig. 1). Witnesses exist exactly when π merges distinct M -classes. The Kovacs sample at its crossing moment and an equilibrated sample at the same volume are the laboratory picture of a witness pair. (SBT: [74], Defs. 2.2–2.4 and Thm. 3.5.)

Memory coordinates and buyback. A witness pair can be resolved by adding coordinates to the lens. The *buyback curve* records, for each budget of added scalars, which declared candidate coordinates resolve the pair. Each addition is a recorded, selected, non-unique choice (a “lift” in SBT terms). Fictive temperature is one such candidate. (SBT: F-II `prop:forgetting-lifts`.)

Four more terms recur. A *lens-definable predicate* is a yes/no function of the readout value; a static order parameter built from the lens is one of these. A *shell* is a frozen set of eligibility conditions under which a certificate is claimed: for GT6, the model and N , the temperature schedule, the observation window, and a non-equilibration criterion, each checked by a membership predicate

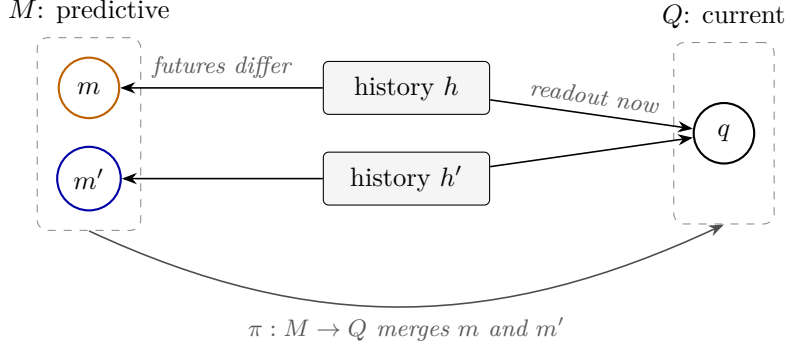


Figure 1. A witness pair. Two preparation histories have the same current readout (the current quotient Q sends both to one class q) but different futures (the predictive quotient M keeps them apart). The comparison map π then merges two predictive classes, and no function of the current readout can predict the future. GT2 exhibits such a pair for the mean energy; GT3 and GT6 for the complete energy distribution, on a constructed pair and on an aging run respectively.

with an exit counter that must read zero. A *strict extension* (SBT: a strict promotion) is the statement that the predictive description M carries information the declared present description Q cannot. At its core is a *nonfactorization* statement: M is not a function of Q . The framework’s fuller filing adds a checklist (a “gate table”) and ablations (a “knockout panel”). Finally, the *six roles* P1–P6 are the framework’s fixed list of ways a description can fail to close. This paper files only P4 (“the future depends on more than the present readout”) and P5 (“the packaging of the present readout is not a closed projection”), and glosses them where they appear.

Claim grades and nonclaims. Every result carries exactly one grade from the framework’s fidelity scale (F-II [71], §9.3). *Theorem-grade (audited class)*: proved or audited unconditionally within the covered finite scope, at the artifact’s declared evidence type. *Certificate-grade (shell-local)*, the standard of paper [C] [76]: valid on the frozen shell, with no broader-class transfer. *Run-only readout*: a measured number with no model fitted to it. *Recognition-grade*: a demonstration rather than a theorem. *Control*: a check that a method returns the known answer. Every artifact carries a machine-readable envelope (`claim_id`, `claim_grade`, `quantifier_domain`, `nonclaims`) validated against a schema in the reproducibility package. Nonclaims are not boilerplate; several are load-bearing, and the full ledger is printed in Appendix B.

2.2 Exact arithmetic and the freeze discipline

Exact rational arithmetic (Python `fractions.Fraction`) carries the kernels, stationary measures, witness gaps, the GT3 predicate sweep, the GT6 full-energy-law witness, the trap-invariance lemma, the GT1 idempotence-defect ratio (289/121), and the lumpable zero control (whose deficit is exactly 0 because exact lumpability is verified symbolically). Quantities that involve logarithms are certified rather than computed in floating point. GT1’s deficits are evaluated from exact joint and conditional probabilities, and every logarithm is enclosed between outward-rounded rational bounds; the artifact declares `evidence_type: exact_rational`, and every inequality of its margin is a statement about those rational logarithm enclosures. The GT6 pressure bounds use the same device for exponentials and logarithms. Each artifact declares its evidence type, and a grade rests on the audited-class quantification plus that declared evidence type. Floating point (`float64_deterministic`) appears only in explicitly labeled diagnostics: printed decimal values, spectral gaps, windowed deficit diagnostics, GT7 Monte Carlo spot checks, the finite pressure ladder and the GB1 three-state

demonstration. Controls accompany every certificate, including the constructed negative controls of the triage panel (§6).

The paper’s evidence base is frozen by sha256 content hashes over canonical JSON. Each freeze artifact carries a `content_hash`, and every cross-artifact citation pins the target’s hash, so regenerating any artifact invalidates, detectably, every citation of it. A repository-wide test suite enforces these bindings. The paper’s own claim citations, the bracketed row labels such as [GT1], are pinned the same way: each carries a hidden artifact hash that is checked against the claims ledger and the current artifact bytes at build time. Content hashes establish *which* bytes were frozen; they do not, by themselves, establish *when* (§8).

2.3 What the reader can check

The claims ledger (`paper/claims_ledger.json`) is the paper’s skeleton: one row per claim with its final wording, grade, quantifier domain, anchors into the SBT corpus, artifact hashes and nonclaims. Each results section transcribes its ledger rows into prose; a discrepancy between prose and ledger is a bug in the prose. A reader with the repository can re-run every certificate from a clean checkout: the test suite, the Lean build with its per-theorem axiom audit, and the three checkers (claims ledger, GT6 filing, empirical bridges).

3 Models and Pipeline

3.1 Model families

Four model families populate the audited class. All are finite-state discrete-time Markov chains with exactly rational kernel entries. Two, East and FA, are kinetically constrained spin models, whose glassy phenomenology comes from constraints on the dynamics rather than from their trivial product statics [30, 20]; the others are an unconstrained spin control and a trap family. Temperature never enters as a free real parameter. Instead we declare a finite rational grid in the Boltzmann activation variable $\epsilon \in \{1/10, 1/5, 3/10, 2/5, 1/2, 3/5, 7/10\}$, where ϵ plays the role of $e^{-\beta}$; the nominal temperature $T = -1/\ln \epsilon$ is an annotation for the reader and never a computational input. The up-spin equilibrium concentration $c = \epsilon/(1 + \epsilon)$ is exactly rational. A temperature protocol is a finite word over this grid, and the position in the protocol is made part of the state, so that the chain stays autonomous and time-homogeneous even while the temperature schedule runs (the clock-audit discipline of F-I [70]).

East model (primary substrate). N binary spins with a frozen up-spin wall at site 0; spin i may flip only if its left neighbor is up ($C_i(n) = n_{i-1}$) [35], the one-dimensional realization of hierarchically constrained dynamics [51]. One step of the random-scan heat-bath (Glauber) kernel K_ϵ [32]: pick a site uniformly; if unconstrained, resample it from the Bernoulli(c) marginal. Because the constraint does not involve the flipping spin itself, K_ϵ is reversible with respect to the product measure $\pi_\epsilon(n) \propto \epsilon^{E(n)}$ with $E(n) = \sum_i n_i$, verified programmatically as exact Fraction identities for every ϵ on the grid. One point matters later: K_ϵ and $K_{\epsilon'}$ are reversible with respect to *different* measures. A temperature protocol therefore violates the common-measure hypothesis of the published protocol-trap theorem, which is what GB1 (§6) addresses.

Fredrickson–Andersen (FA-1f). Identical structure with the symmetric constraint $C_i(n) = \max(n_{i-1}, n_{i+1})$ [29]. FA is the transfer target for GT7. FA hump outcomes are not used to construct the predictions; the FA kernel’s spectral gaps enter the declared interval-rescaling rule. The declared

protocol places the FA target runs after prediction freezing; the retained content hashes do not independently certify that order (§8).

Unconstrained control. The same heat-bath chain with $C_i \equiv 1$. Here the energy readout is lumpable at every horizon (each site resamples independently, so the number of up spins is itself a Markov chain), and the closure deficit is exactly 0 as a rational number. This is the zero control for GT1 (§4).

Bouchaud-type trap model (second transfer target). M traps with declared rational escape weights w_k ; from trap k , jump to a uniformly random trap with probability proportional to w_k , else stay [28, 12, 44]. Temperature dependence is a declared finite family of rational weight vectors, one per grid ϵ (exponentiating weights would break rationality); the physical Arrhenius relation is again an annotation, not an input. The stationary measure is $\pi_k \propto 1/w_k$, verified exactly.

3.2 A proven limitation of the declared trap family

[FINDING.trap_epsilon_invariance] The declared trap weight family multiplies every trap’s base weight by a common per- ϵ scalar. Since $\pi_k \propto 1/w_k$ depends only on weight *ratios*, that scalar cancels exactly: the trap stationary distribution is ϵ -invariant (proved in exact Fraction arithmetic; regression-tested). The declared temperature action therefore governs relaxation *rate* only, never the relaxation *target*. This is known in substance in the trap-model literature [12, 44, 57, 6, 27, 9]: the standard convention is a per-trap Arrhenius factor $w_k(T) \propto e^{-E_k/T}$, under which different traps rescale by different temperature-dependent factors and the equilibrium genuinely shifts, which is exactly the mechanism used to produce Kovacs memory in trap models [27, 9]. Our multiplicative-scalar choice was made deliberately to keep every computation exactly rational; the invariance is its elementary, proven consequence. It is filed here as a design caveat because it has a concrete downstream cost, paid in §8: a Kovacs protocol started from equilibrium cannot start the trap family below its post-jump equilibrium, so the trap column of the transfer scoreboard is structurally unobservable. This is a property of our declared family, not a no-go theorem for trap models.

Nonclaim: Not a novelty claim: the cancellation is standard trap-model structure once stated; the contribution is stating, proving, and paying for it within the declared family. It says nothing about trap dynamics or aging trajectories; only the stationary distribution is ϵ -invariant.

3.3 Pipeline

The exact-finite pipeline ports paper [A]’s package specification [74] to thermal semantics: rational history distributions over a declared preparation catalog, rational row-stochastic continuation kernels for each protocol word, and two-time observables as future-signature entries. Parity with [A]’s benchmark family (a six-regime control suite, re-instantiated with thermal semantics) is filed as a control artifact and consumed by GT4 (§6). Spectral gaps of K_ϵ (float64 diagnostics, never certificate content) calibrate the τ windows and aging times used throughout (Fig. 2). They and the derived “mixing-time” field are calibration proxies, not proved total-variation mixing bounds. Their rapid closing as ϵ decreases is the East model’s super-Arrhenius relaxation, established numerically and rigorously in [66, 2, 18].

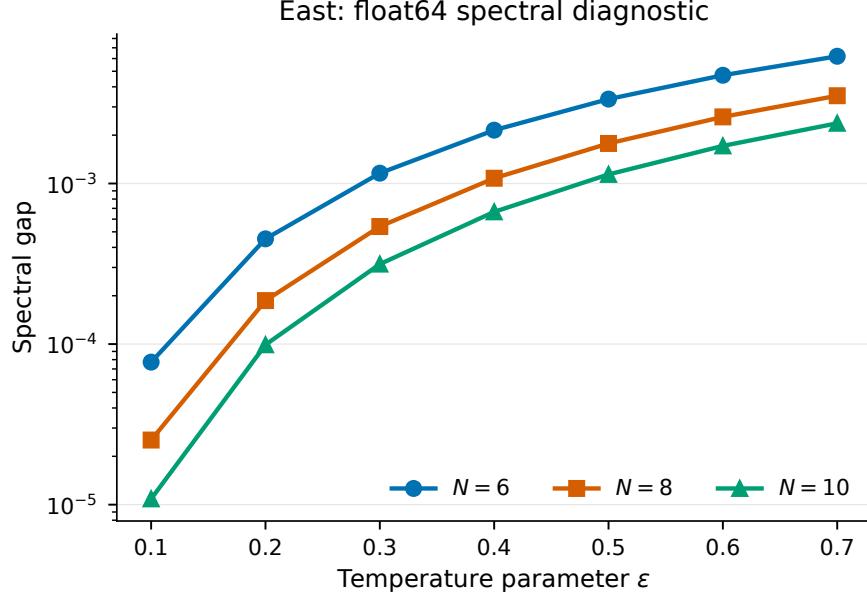


Figure 2. East-model spectral gaps across the declared (N, ϵ) grid (float64 diagnostic used only for window calibration).

4 GT1: The Energy Readout Is Not Closed

4.1 Claim

In plain terms: for an aging East chain, the microstate tells you substantially more about the energy a few steps ahead than the present energy does, and it does so by a certified factor more than at constrained equilibrium. Precisely:

[GT1] On the declared scope (East model, $N = 8$, lens L_{energy} , horizon ladder $\tau \in \{1, 2, 4, 8\}$, quench $\epsilon : 7/10 \rightarrow 1/10$ with waiting time $t_w = 20$), the closure deficit CD_τ of the aging protocol exceeds the constrained-East stationary-equilibrium baseline at every declared horizon by at least the declared margin $3/2$. The certified lower bound on the smallest ratio is 1.6391, at $\tau = 8$ (Table 2). Every inequality is proved in exact arithmetic: the joint and conditional probabilities are exact rationals and each logarithm is enclosed between outward-rounded rational bounds, so the decimal values shown are illustrative. Alongside, the one-step idempotence defect of the packaging map is $1225/1936$ for the cold ($\epsilon = 1/10$) quench kernel and $1225/4624$ for the hot ($\epsilon = 7/10$) equilibrium kernel, an exact ratio $289/121$ above its declared margin 2. Grade: theorem (audited class), exact rational. In the framework’s filing vocabulary: the P5 channel of the standard thermodynamic packaging is not **active_projection** as a closed packaging on this scope; packaging the present energy readout does not yield a closed description. Anchors: F-III T8, F-I D-IC-01/D-IC-02/T-CL-01, [75] Def. 3.1.

Table 2. GT1 closure deficit (nats) of the aging protocol against the constrained-equilibrium baseline (East $N = 8$, energy readout). Deficits are shown as floats; the last column is a rigorous lower bound on the ratio from exact probabilities and rational logarithm enclosures, rounded down

τ	CD_τ (aging)	CD_τ (equilibrium)	ratio	certified ratio
1	0.033650	0.018875	1.7828	1.7828
2	0.063730	0.036068	1.7669	1.7669

τ	CD_τ (aging)	CD_τ (equilibrium)	ratio	certified ratio
4	0.113110	0.065575	1.7249	1.7248
8	0.177212	0.108112	1.6392	1.6391
margin	–	–	declared 3/2	≥ 1.6391

The two packaging defects are each bounded by a different, larger quantity, the prototype-retention error (10/11 cold, 10/17 hot), and the two quantities are reported separately. As stated in §2, the defect is a saturation diagnostic, not a closure criterion; the closure claim rests on the deficit.

4.2 The controls, and why equilibrium is not zero

The controls deserve emphasis, because the intuitive control is wrong. One might expect an equilibrated chain to have nothing left to remember, and so to serve as a zero-deficit control. That expectation is *false*, and provably so. $CD_\tau = 0$ for a fully supported distribution if and only if the energy readout is τ -lumpable, and lumpability depends on the kinetic constraint, not on stationarity. In East a spin can flip only when its left neighbor is up, so the number of spins free to move, and hence the rate at which the energy can change, depends on *where* the up spins sit, not merely on how many there are. The energy readout of East is therefore not closed even at equilibrium.

The certificate uses two controls with different jobs (Fig. 3). The *unconstrained* model, whose energy readout is lumpable by construction, sits at exact rational $CD = 0$ (exact lumpability verified symbolically; float evaluation gives residuals near 10^{-17}). Constrained-East equilibrium is an honest *nonzero baseline*. The content of the certificate is that aging exceeds even that baseline by the declared margin. Equilibrium is never used as a zero control.

[GB2] Because the aging chain is driven by a temperature schedule, the closure deficit must be defined relative to the protocol rather than to a single stationary kernel. GB2 supplies that machinery: finite protocol-relative deficit computations (East versus unconstrained, $N = 4$, declared ϵ grid), with exactness reserved for the lumpability check and float64 windowed diagnostics labeled as such. GB2 is recognition-grade, a demonstration rather than a certificate. The GT1 artifact carries a passing consistency gate for it: on a single stationary cell the protocol-relative deficit equals the stationary deficit by definition, with residual exactly 0.

4.3 Scope and an open control

The certificate quantifies over exactly its declared cell: East, $N = 8$, L_{energy} , the declared horizon ladder and the declared quench. FA replication, the richer L_{panel} lens and $N = 10$ are extensions that were not run, listed as such in the artifact envelope.

[NEG.gt1_b_parity_control_open] One intended control is open: the benchmark-parity check against the example chains of paper [B] was not ported, and the GT1 artifact files it as an explicitly skipped control with its reason (machine-readable `status: skipped`). The margins are internal to the declared scope and do not route through [B]’s example chains; no [B]-parity is claimed until the control is run.

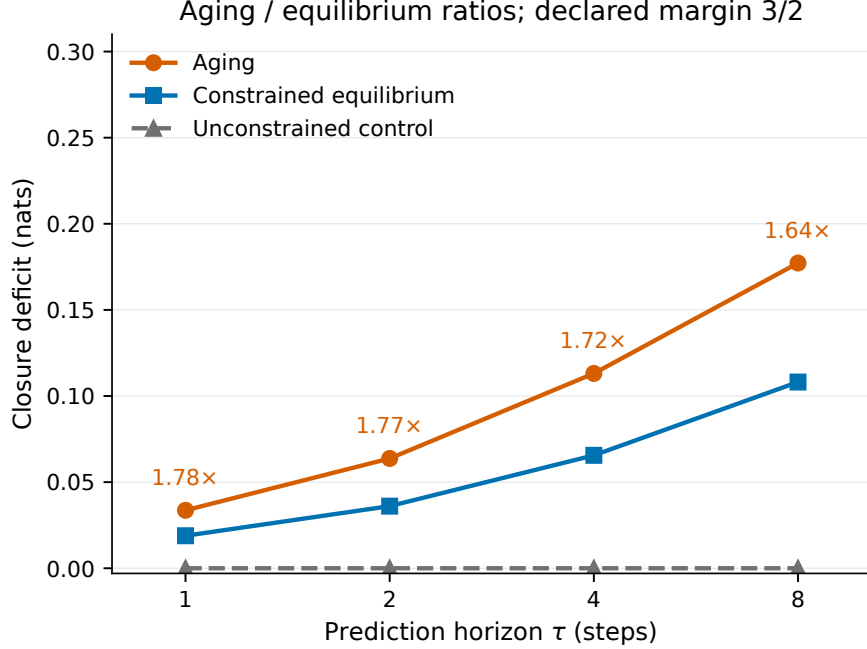


Figure 3. GT1 closure deficit (nats) versus horizon τ : aging protocol, constrained-equilibrium baseline (nonzero because the energy readout is not lumpable), and the unconstrained control at exact zero. Annotations give the aging/equilibrium ratio at each horizon; the declared margin is $3/2$. Plotted values are floats; the margin is certified with rational enclosures (Table 2).

5 GT2 and GT5: The Kovacs Witness and What It Costs to Repair

5.1 The Kovacs protocol on the audited class

The Kovacs protocol [40] is: equilibrate hot, quench cold, age for t_w , jump up to an intermediate temperature, and watch a one-time observable relax. We located it on East by an exact search, run before any certificate and filed as a pre-certificate control. At $N = 10$, with $\epsilon : 7/10 \rightarrow 1/10 \rightarrow 2/5$ and $t_w = 40$, the mean energy after the second jump starts below its new equilibrium value $20/7 \approx 2.857$ (at ≈ 2.741), crosses it between steps 1 and 2 (≈ 2.803 and ≈ 2.859), and overshoots through a hump peaking at $t_{\text{peak}} = 35$ before returning (Fig. 4), all in exact rational arithmetic. We write $t_K = 1$ for the last step before the crossing. The raw trace never equals $20/7$ at an integer step; the preparation that does, a randomized stop between steps 1 and 2, is constructed in §5.2.

The non-monotone excursion after the crossing is the classic signature that the present value does not determine the future. For a finite Markov chain in the linear-response regime, Prados and Brey showed that it is generic whenever the observable’s relaxation is not a single exponential [55]. That East- and FA-type models show a Kovacs hump was established numerically by [16, 4]; what is new here is the exact-rational trace and the certificate built on it.

5.2 GT2: the witness certificate

In plain terms: we exhibit two preparation histories of the East chain with exactly the same present mean energy and provably different mean energies five steps later.

The first history is the Kovacs run stopped at the crossing. Since no single time step lands exactly on $20/7$, the run is stopped at step 1 with probability $\theta \approx 0.0359$ and at step 2 otherwise, with θ solved exactly so that the mixture’s mean energy is $20/7$ (a “randomized stop”, a declared

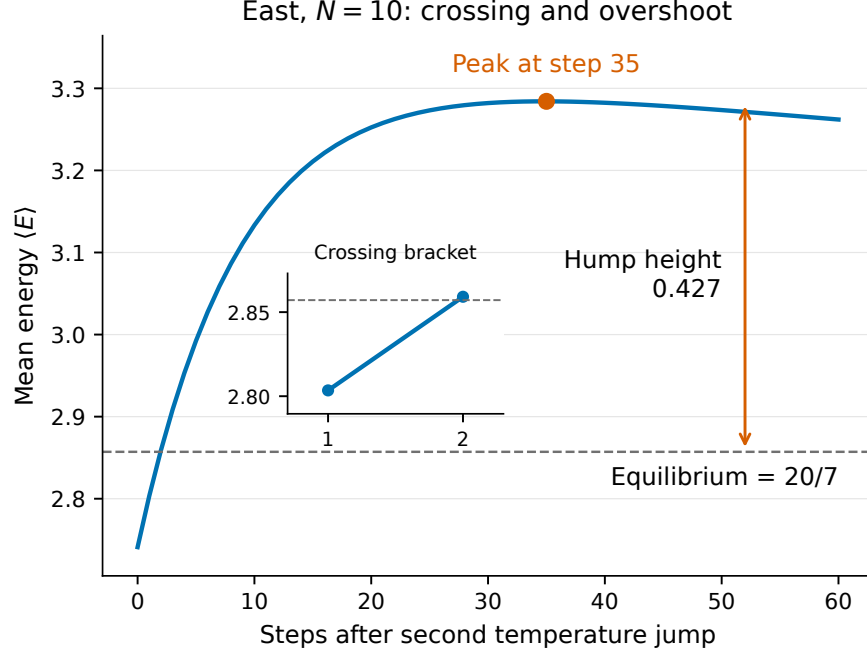


Figure 4. Kovacs hump on East $N = 10$ (exact-rational trace, floats only for plotting): the mean energy starts below its equilibrium value $20/7$, crosses it between steps 1 and 2 (inset), peaks at step 35, and returns toward equilibrium.

move of the pipeline). The second history is the equilibrium distribution at $\epsilon = 2/5$, whose mean energy is $20/7$ by definition. Both are then held at $\epsilon = 2/5$ for five steps. The equilibrium history stays at $20/7 \approx 2.857$; the Kovacs history moves to ≈ 3.058 , a gap of ≈ 0.2005 . Only the mean energy is matched at the start, not the full energy distribution, and that is exactly what this lens sees. (GT6 removes this limitation: there, the whole distribution is matched.) Precisely:

[GT2] On the declared scope (East, $N = 10$, `kovacs_moment` lens, declared ϵ grid and continuation catalog), one exact-rational split pair of preparation histories is exhibited: the two histories agree on the present lens value (mean energy $20/7$) and their futures differ, with a strictly positive exact rational maximum future gap (≈ 0.2005 on the five-step hold; the exact fraction is in the artifact). By [74] Thm. 3.5, such a pair exists if and only if the comparison map $\pi : M \rightarrow Q$ is strictly refining: the predictive description is genuinely finer than the present one. The exported rational table of this pair is also checked in Lean (§9). Grade: theorem (audited class).

A “different future” could mean three things, and we claim only the first: that the future depends on more than the present readout. We do *not* claim that the two histories took different routes to the same state (a P3 route mismatch), and we claim no directionality or irreversibility. In the framework’s vocabulary the claim is **P4 staged-dependence**, filed on the standard witness template (F-II `prop:p1-witness`). No P6_drive audit was performed, so no arrow-of-time statement of any kind is made on GT2’s behalf.

[NEG.gt2_loop_certificate_not_built] A companion certificate about thermal loops was also planned: a closed temperature cycle that acts trivially on the present description Q but nontrivially on the predictive description M . It is not filed. A tuned real East loop does return the current mean exactly, but the declared family of histories it generates does not close within six rounds of iteration (it reaches 14 histories), so no action of the loop on the quotients is certified. A constant-temperature control loop scores zero on both quotients, as it must. The split-pair witness stands alone, and nothing downstream depends on the loop.

5.3 GT5: what it costs to buy the distinction back

In plain terms: adding a single scalar to the energy readout is enough to tell the two GT2 histories apart, and a fictive-temperature-like index is one of three declared choices that work. Precisely:

[GT5] On the declared scope (East, $N = 10$, `kovacs_moment` interface, the declared two-history class), the present lens collapses the witness pair into one class ($|Q| = 1$ on the pair) while the predictive quotient keeps two ($|M| = 2$), so the largest fiber of π has size $\text{MaxFiber} = 2$, meeting the nontriviality threshold. The buyback curve (Fig. 5) is published as a lift ledger per F-II `prop:forgetting-lifts`: each added memory coordinate is a recorded, selected, non-unique lift. At budget 0 the pair is unresolved. At budget 1, *each* of the three declared scalar coordinates already resolves it exactly (loss 0), so the shipped witness saturates at `saturation_budget` = 1. Grade: theorem (audited class).

The three coordinates are expectations, over the history’s microstate distribution μ , of a per-microstate quantity:

- $T_f(\mu) = \mathbb{E}_\mu[i^*(E(n))]$, where $i^*(E)$ is the index of the grid value ϵ_i whose equilibrium mean energy is closest to E . This is an index into the declared grid, not a temperature. It is computed microstate by microstate, not from the history’s mean energy, so two histories with equal mean energy can have different T_f .
- $D(\mu)$, the expected number of adjacent up-up spin pairs.
- $n_1(\mu)$, the expected occupation of the site next to the wall.

The loss is the sum of the two histories’ squared errors when both are predicted by their common mean, with unit weights; it is an in-sample bookkeeping quantity, not a held-out generalization error.

The comparison with the GT6 pair (§7) is instructive. T_f depends on the history only through its energy distribution. On the GT2 pair only the mean energy is matched, so T_f can and does separate it. On the GT6 pair the entire energy distribution is matched, so the T_f difference is exactly 0 there, as it must be for any descriptor computed from the energy law, while D or n_1 alone still resolves that pair at budget 1.

The translation to the glass literature is deliberately modest. The TNM fictive temperature [69, 49, 47, 33] is the motivating analogy for T_f : an energy-indexed effective temperature, added as a declared step from the present description toward the predictive one. The implemented T_f is the grid-index construction above, not the TNM quantity itself. Modern treatments of the Kovacs effect likewise pair an effective temperature with a second structural variable [14], and the multiparameter models the Kovacs literature moved to [41] are why the general question is left open here rather than answered on two pairs. The certificate demonstrates the repair *mechanism* and its ledger discipline. $\text{MaxFiber} = 2$ does not bound how many real-valued coordinates memory needs in general, and we do not claim that a single fictive-temperature-like scalar is insufficient in general.

6 GT3 and GT4: Foreclosure and the Kill-Test Panel

6.1 GT3: the order-parameter search, foreclosed on its declared class

In plain terms: a static order parameter built from the energy readout is a function of the energy value. On $N = 8$ spins the energy takes 9 values, so there are exactly $2^9 = 512$ yes/no functions of it, and we can check them all. Every one of them agrees on a pair of histories that a finer readout of the same microstate, the occupation of the site next to the wall, tells apart. Precisely:

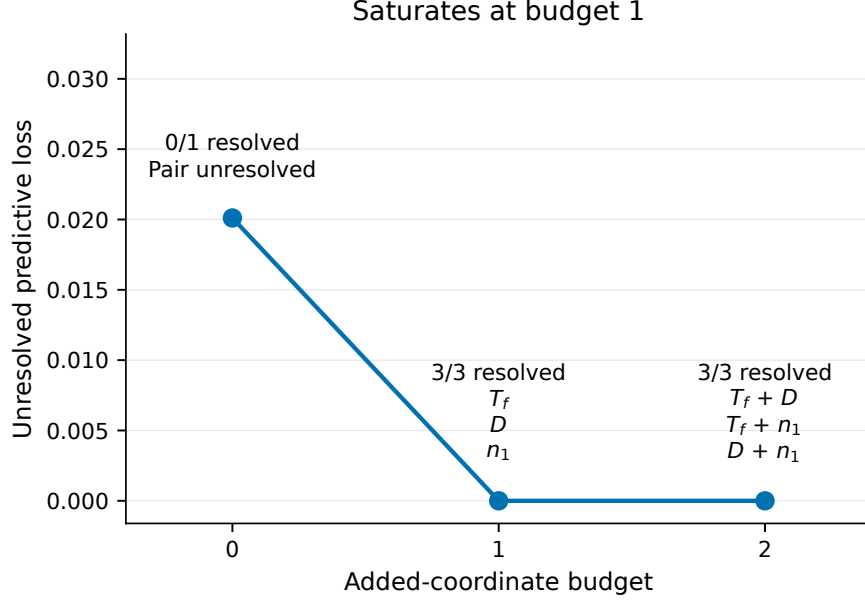


Figure 5. GT5 buyback curve for the GT2 pair: unresolved predictive loss versus added-coordinate budget, annotated with resolved/attempted counts and the resolving coordinates. The loss is the sum of the two unit-weight squared errors under common-mean prediction, $(a - b)^2/2$, where a and b are the two histories' five-step future mean energies. It is incurred only while the pair is left in one class (≈ 0.0201 at budget 0, half the squared witness gap); a coordinate that separates the pair has loss 0. The witness saturates at budget 1.

[GT3] On the declared scope (East, $N = 8$, $\epsilon = 1/2$), the definability bound of F-III T14 makes the static-order-parameter search space finite: a fixed lens with image size 9 defines exactly $2^9 = 512$ predicates. The sweep checks all 512. Every one agrees on the exhibited witness pair, while a readout based on n_1 (the occupation of the site next to the wall) separates it, with exact gap $169/1536$ under the identity continuation (Table 3). The witness pair is a standalone reflection pair constructed for this sweep and deliberately *not* chained to the GT2 certificate, so the two results fail independently. The foreclosure is constructive: the distinguishing readout is exhibited and proven absent from the set of energy-definable predicates, not inferred from counting. Grade: theorem (audited class).

Table 3. GT3 foreclosure sweep summary (East $N=8$, $\epsilon = 1/2$)

field	value
N	8
epsilon	1/2
lens	L_energy
total predicates	512
agreeing predicates	512
lens image size	9
all agree	True
separating event	n_1

field	value
witness provenance	standalone W_reflection witness constructed for this sweep; not chained to the GT2.kovacs_moment certificate (a separate witness family per design/specs/04_gt2_witnesses.md §4).
T-FOR-02 statement	the distinguishing readout (n_1-based) is not in Def(l_energy); constructive witness exhibited, not generic counting

The field-facing sentence “there is no static order parameter for glass” is licensed here *only* as the declared-class shadow of this theorem (F-II item E2), with the class, the catalog and the suppressed content named in the same breath: no predicate definable from the declared energy lens on the declared class separates what memory separates. Reading it as a statement about other instruments, other lens classes (in particular the amorphous order of random first-order transition theory [39, 42], the configurational-entropy line descending from [1], or the local-structure measures surveyed in [61]), or the continuum is not licensed, and this paper does not do so.

What the theorem does license, via the framework’s no-free-distinction law (F-IV F12), is narrower and exact: any readout that *does* separate this pair must carry information beyond the energy readout. Here that information is a present structural coordinate of the microstate (n_1 , under the identity continuation), not a record of the history. The separation shows that the energy-based description is incomplete. In fact the two histories have the same complete energy distribution, so every descriptor computed from it, not only the 512 predicates, agrees on them. GT6 (§7) supplies the same full-law match for two preparations of a real aging run, separated by the East dynamics rather than by a present readout, and GT5 shows that one added coordinate suffices to repair the Kovacs pair.

6.2 GT4: the kill-test panel

A memory signature can be faked. GT4 builds five schematic fixtures for the ways it can be faked and checks that the same deterministic classifier that accepts the Kovacs candidate rejects all five. The candidate, `kovacs_wheel`, is a reflection pair at $N = 8$ that the full spin panel distinguishes but its reflection-invariant part does not. The control fixtures use the 16-state $N = 4$ carrier and two or three interfaces, with declared readouts and continuations; they are regime schemata, structural toys, not physical mechanisms:

- The *protocol trap*: a pair whose readout difference is a residue of its schedule. Its cleared twin exposes the toy clock label in the current readout while leaving every future unchanged, so the pair is visibly different now and the apparent memory disappears.
- The *flattenable* pair: an apparent memory caused by an incomplete continuation catalog. Its cleared twin applies an admissible completion that averages the declared two-state orbit, an idempotent stochastic map on the same support, after which the futures agree.
- The *explicitly latent* pair differs in a bit that the present readout does not see.
- The *dissipative* fixture merges the two histories at a later interface.
- The *flat* control is two copies of the $N = 4$ equilibrium distribution, which by construction has nothing to separate.

Precisely:

[GT4] Over the declared 9-member benchmark family of small structural packages, the deterministic exact triage classification identifies `kovacs_wheel`, the Kovacs memory candidate, as the sole `coherent_candidate`, while the five declared non-coherent regime controls (`artifact_trap`, `dissipative`, `explicit_latent`, `flat`, `flattenable`) are correctly killed or cleared (Table 4). Protocol-trap discipline is applied: raw and cleared classifications are both recorded, so the table shows each artifact being caught. For instance, the naive protocol-trap fixture classifies as `artifact_trap` raw, and its clock-exposing twin as `flat`. Grade: theorem (audited class), over the declared finite family only, not a shell-general theorem. The robustness fractions cover three declared structural perturbations; they are not probabilistic confidence guarantees.

Table 4. GT4 regime triage over the declared benchmark family: each fixture and the class the deterministic triage assigns to it, before (raw) and after (cleared) the declared repair

control	raw fixture	raw class	cleared fixture	cleared class
coherent candidate	kovacs_wheel	coherent_candidate	–	–
artifact_trap	protocol_trap_naive	artifact_trap	protocol_trap_honest	flat
dissipative	dissipative_memory@mid	dissipative	dissipative_memory@end	flat
explicit_latent	latent_memory_base	explicit_latent	latent_memory_refined	flat
flat	flat_control	flat	(pure control)	–
flattenable	flattenable_raw	flattenable	flattenable_completed	flat

6.3 GB1: the protocol-trap caveat, carried openly

The “protocol trap” is a warning about arrows of time, not about memory. A system whose dynamics switch between several kernels under a clock can show a spurious time asymmetry if the clock is left outside the state. The framework’s published theorem on this (F-I T-AOT-02) concerns the random-scan autonomous lift initialized at its stationary product law $\pi \times s$: when the clock is reversible and every phase kernel is reversible with respect to a *common* measure π , that lift, started stationary, has zero entropy production and zero path-reversal asymmetry at every horizon, so any observed arrow must come from an external schedule, a non-stationary start, or a stroboscopic fit. The conclusion does not cover arbitrary initial laws. Kovacs kernels K_ϵ are reversible with respect to a different Gibbs measure for each ϵ , so the theorem does not apply as published: the temperature switches contribute a boundary term.

[GB1] GB1 is the temperature-protocol version of that audit, in the style of the protocol-trap microarticle [78], at recognition grade. Three facts are established. (i) An exact identity expresses the stationary entropy production through a candidate measure built from the phase Gibbs laws plus a boundary term. (ii) That candidate expression is *not* a general upper bound: an exact rational three-state counterexample, with a reversible clock and a common energy function, exceeds it by more than 9.5×10^{-8} , certified with rational logarithm bounds. (iii) A corrected stationary budget is proved: it adds an explicit residual, $\text{osc}(b) k \text{TV}(\hat{\mu}P, \hat{\mu})/(1 - D_k)$, controlled by a Dobrushin contraction coefficient $D_k < 1$, and it recovers the published zero-entropy-production

conclusion when all phases share one Gibbs law and the clock is reversible. No bound of this form holds for arbitrary initial laws; a two-state transient counterexample shows why. The three-state demonstration in the artifact is a float64 numerical check.

GB1’s role is to keep the arrow-of-time audit separate from the memory classification. GT4’s triage computation is deterministic and exact, but the clock-audit *discipline* it applies (raw versus cleared classification) rests, for temperature protocols, on a recognition-grade demonstration with a stationary budget rather than on a closed transient theorem. This is the framework’s conditional-reporting rule working as designed.

7 GT6: The Memory Layer Strictly Extends the Full Energy Law

GT2 matches only the mean energy. GT3 already matches the complete energy distribution, but on a constructed reflection pair separated by a present structural readout. GT6 asks for the same full-law match in real aging: two preparations of *one aging run* that agree on the complete current energy distribution, and therefore on every quantity computed from it, yet are driven apart by the East dynamics.

7.1 The construction, in plain terms

The shell is East with $N = 8$ spins and a frozen up-spin wall. Start in equilibrium at $\epsilon = 7/10$, quench for 20 steps at $\epsilon = 1/10$, then switch to $\epsilon = 2/5$. Let μ_t be the microstate distribution after t further steps, for $t = 0, \dots, 9$, and let a_t be its energy distribution, a vector of 9 probabilities. Ten vectors in a nine-dimensional space are linearly dependent, so there are coefficients c_t , not all zero, with $\sum_t c_t a_t = 0$; because each a_t sums to one, the coefficients sum to zero. Splitting c into its positive and negative parts and normalizing each gives two probability distributions over stopping times, p and q . The two preparations are randomized stops of the same run: h_+ stops at time t with probability p_t , and h_- with probability q_t , each chosen independently of the spin trajectory. By construction,

$$\sum_t p_t a_t = \sum_t q_t a_t,$$

so h_+ and h_- have *identical* current energy distributions. The selector uses only these current energy vectors; no future readout or equilibrium target enters it. On the frozen data the ten vectors have rank 9, and the null vector puts h_+ on the odd stopping times and h_- on the even ones (Fig. 6a). Both preparations are then continued at $\epsilon = 2/5$ and their mean energies compared at every horizon $\tau = 1, \dots, 21$; with $9 + 21 = 30$, every operation stays inside the declared observation window.

7.2 The claim

[GT6] On the audited East shell, the two randomized-stopping preparations h_+ and h_- have exactly equal current energy distributions, while their future mean energies differ at every horizon $\tau = 1, \dots, 21$ (exact rational gaps; the smallest, at $\tau = 1$, is $\approx 1.49 \times 10^{-10}$, and they grow to $\approx 1.13 \times 10^{-9}$). By F-III T12/T13, a split pair of this kind implies that the predictive object map does not factor through the complete current energy law: the object-map verdict is **strict**. Consequently all 512 energy predicates, and every descriptor computed from the current energy law, including the energy-derived T_f of GT5, fail to be future-sufficient on this pair. Replacing only the probe kernel by the unconstrained heat bath, whose energy process is lumpable, makes every one of the 21 future gaps exactly 0. The exported rational tables are checked in Lean (**EnergyN8**, §9); the

preparation, the null-vector selector and the evolution are exact Python. Grade: certificate (shell-local). Paper [C]’s fuller `thm:strict-extension` is not invoked, and the framework’s verdict-level theorems T19/T20 are not used to supply any missing gate.

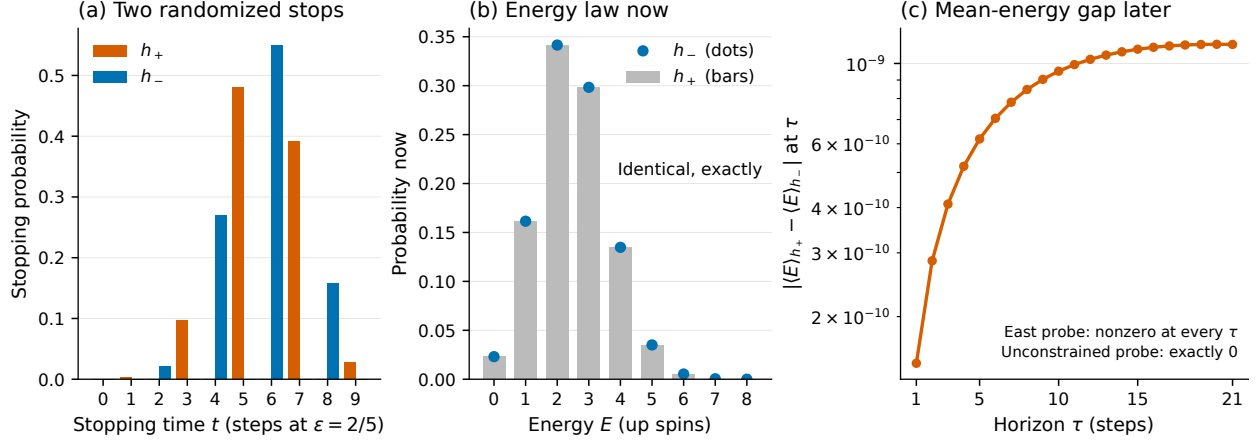


Figure 6. The GT6 full-energy-law witness (East $N = 8$; exact rationals, floats only for plotting). (a) Stopping-time distributions of the two preparations, read off the positive and negative parts of one null vector of the current energy distributions. (b) Their current energy distributions coincide exactly. (c) Their future mean energies differ at every horizon from 1 to 21; the gaps are small but exactly nonzero. With the probe kernel replaced by the unconstrained heat bath, every gap is exactly zero.

Three remarks fix how far this reaches. First, the logic is elementary. If some function F mapped the current energy law to the future, equal current laws would force equal futures, and the certificate exhibits equal current laws with unequal futures. The exactness of the arithmetic is what makes a 10^{-10} gap a proof rather than a rounding artifact. Second, the gap is tiny, and we claim nothing about its experimental detectability. A total-variation argument gives an explicit, equally tiny, robustness margin: perturbing each source law by at most $d/32$, with d the smallest gap, keeps every future gap above $d/2$. Nonfactorization survives such a perturbation only if the perturbed preparations still have equal current energy laws. Third, the two preparations are independent random stopping mixtures, an explicit extension of the deterministic-run catalog; that extension is declared, not hidden.

The ablation is the control that makes the result about the East constraint. The unconstrained heat bath at $\epsilon = 2/5$ moves the energy by one unit with probabilities that depend only on the energy itself ($((8 - j)(2/7))/8$ up and $j(5/7)/8$ down at energy j), so equal energy laws stay equal forever. The separation in Fig. 6c therefore comes from the kinetic constraint, through the arrangement of up spins that the energy law does not record.

7.3 The fuller filing, and why it does not pass

The framework’s complete strict-extension filing asks for more than nonfactorization. Its gate table (Table 5) includes a stability gate, G_{stab} , which requires the packaging map $E_{\tau,f}$ of §2 to have multiple stable fixed-point strata, one per persistent memory class.

[NEG.gt6_original_filing_failed] On this shell that gate fails, and provably cannot be repaired by tighter numerics. At $\epsilon = 2/5$ and $\tau = 1$, the packaging kernel is irreducible (an exact graph check) and preserves the Gibbs law; a finite irreducible stochastic kernel has exactly one stationary probability law, so there are no multiple fixed-point strata to find. Its maximum prototype-retention error is $5/7$, far from small. That the set of reached support labels saturates does not certify

fixed points. The full filing verdict is therefore **failed**, while the object-map verdict above stands. Restoring the full filing would need a different completion or limit construction with its own applicability proof; the finite-window persistence of Fig. 6c is not substituted for the missing gate. Semantic exclusions in the admissibility domain are recorded as pending rather than assumed clear.

Table 5. GT6 gate table of the original strict-extension filing, as filed in the audit artifact

gate	required	status	evidence
G_suff	yes	pass	actual current-retaining finite maps
G_desc	no	not_required	glass bridge claims strict object-map enrichment, not closed macro dynamics
G_stab	yes	failed	requires packaging fixed points; support-label saturation is insufficient
G_ctrl	yes	pass	same-source unconstrained probe ablation
G_nosmuggle	yes	pass	current-only convex selector
G_vis	yes	pass	actual citations and scoped nonclaims
G_audit	yes	pass	exact certificate source replay
G_strict	yes	strict_pass	same-shell full-energy-law split pair; F-III T12/T13
G_logglob	no	not_required	shell-bound single bridge only; no stacked or glued promotions claimed

The knockout panel (Table 6) disables, in turn, each of the six primitives P1–P6 that the framework says a closed description can depend on, and records the effect in two distinct modes. For P1–P3 the knockouts are *numeric*: the primitive is replaced (identity run kernels; the unconstrained lumpable kernel; constant ϵ) and a declared numeric diagnostic degrades past its declared threshold. For P4–P6 they are *capability-loss* rows: removing the primitive makes a downstream diagnostic unavailable rather than numerically degraded, and no numeric threshold is claimed. Two rows use auxiliary systems: P5 uses an $N = 8$ packaging diagnostic, and P6 uses a separate three-state system with a two-phase clock. Because of this, the panel is a scoped diagnostic survey. It does not show that six primitives act causally in one glass dynamics, and no full-loop lawfulness is claimed.

Table 6. GT6 knockout panel (P1–P3 numeric ablations; P4–P6 capability-loss rows)

primitive	kind	degraded	threshold
P1	numeric	True	identity trace variation must be exactly 0
P2	numeric	True	unconstrained control must be exactly lumpable under L_energy
P3	numeric	True	constant-epsilon replacement must fail the hump detector
P4	capability_loss	True	–
P5	capability_loss	True	–
P6	capability_loss	True	–

[NEG.material_forcing_absent] One more route is also closed. Paper [C]’s strict-extension theorem needs *material forcing*: the protocol catalog must contain conditioned schedules, in which the next temperature step depends on the measured panel, as in real glass processing, and some distinctions between histories must exist only because of them. The audited catalog has `material_forcing_count = 0`, so that theorem is not invoked anywhere. This marks the boundary of the [C] machinery this instantiation exercises; it is not evidence about other shells or protocols.

7.4 Pressure: what can and cannot be said

The filing also carries a thermodynamic-pressure diagnostic: tilt the transition kernel by e^{sE} and measure the exponential growth rate of its powers. One might hope that conditioning on energy fibers would lower this pressure by a positive amount, a “conditional gap”. The exact analysis settles which versions of that hope survive.

[FINDING.gt6_pressure_limits] For the tilted East operator on the shell ($N = 8$, $\epsilon = 2/5$, tilt $s = 1/5$), positive comparison vectors and exact exponential and logarithm enclosures bound the limiting (Fekete) pressure to $[1.039322, 1.039618]$. Because some power of the tilted kernel is strictly positive, *every* nonzero initial law has this same limiting pressure. Conditioning only the initial law on an energy fiber, and then running the original dynamics, therefore gives an infinite-time weighted gap of exactly zero. Two positive statements remain, each with its own meaning. With the original dynamics, the finite-time Jensen contrast is certified to be at least $1/1000$ for every horizon $n = 1, \dots, 8$; but it is already present for the stationary comparison law, so it is not a glass-specific witness. And killing paths when they leave an energy fiber gives a survival-pressure contrast certified above $1/2$ (≈ 0.6795); killing changes the dynamics and is not initial-law conditioning. Grade: theorem (audited class) for the stated operator. The finite pressure ladder in the filing’s support table is a float diagnostic, not a limiting pressure.

7.5 What strictness does not mean

The nonclaims (Appendix B) are part of the result. Strictness here is shell-local and finite-window: no shell-general strict-bridge theorem, no all-time persistence, no large- N statement. It does not imply that the memory layer has an autonomous closed dynamics of its own, nor that anything drives it (no P6_drive audit was performed). The pressure quantities are neither closure deficits nor the strictness witness; the energy-law split pair is. And the promotion is a single bridge $T_0 \rightarrow T_1$; the framework has no theorem for stacking such bridges, and none is used.

8 GT7: Aging Constants Are Run-Only: Freeze, Transfer, and the Scoreboard

8.1 The protocol

GT7 asks whether the numbers that characterize aging on East (the Kovacs crossing time, the peak time, the hump height, and three others) carry over to a different model without any fitting. The protocol follows the logic of a pre-registered experiment [50, 48]: fix the readouts, compute the East reference, commit predictions for the target models, and only then run the targets and score them with a frozen script. Precisely:

[GT7] GT7’s claim is a discipline plus its unedited outcome. The protocol has five stages, each frozen by an sha256 content hash:

- **F1, readouts freeze.** Six declared readouts (Kovacs t_K , t_{peak} , hump height, memory-dimension profile, deficit-decay slope, cross-validation checks), frozen in a registry with content hash and freeze note.
- **F2, East reference.** The reference surface on East: 56 exact hump-surface cells at $N = 10$, an $N = 16$ Monte Carlo spot check, the closure-deficit ladder (Fig. 7) whose declared secant slope is the frozen deficit-decay readout, and memory-dimension profiles. Run-only readouts; no functional form is fitted anywhere.
- **F3, pre-registration.** 338 prediction cells: 42 interval predictions, 2 order-relation predictions, and 294 cells explicitly declared `no_prediction`. The predictions registry pins the hashes of the frozen readouts, the East reference and the scoring script. The scored predictions concern the three hump readouts (t_K , t_{peak} , height); the memory-dimension and deficit-decay readouts are not transferred. East intervals are rescaled to each target by a declared rule that uses the target kernel’s spectral calibration, never its hump outcomes.
- **F4, target runs.** The FA and trap families are run against the registered predictions.
- **F5, mechanical scoring.** The frozen scorer (hash-pinned at F3, verified unchanged) compares predictions with observations, with no post-hoc widening or reinterpretation.

What the hashes do and do not establish should be stated plainly. They bind the scored predictions to the exact bytes of the readouts, the reference and the scorer, so no prediction or scoring rule can have been altered without detection. They do not, by themselves, certify the chronological order of the stages, and the readouts registry’s own freeze note records that the planned per-stage commits were not made. “Run-only” is a procedural grade: no same-present, different-readout witness is claimed for each individual GT7 readout. (GT3’s non-definability result concerns its own witness pair and the energy lens; it is not extended to the GT7 readouts.)

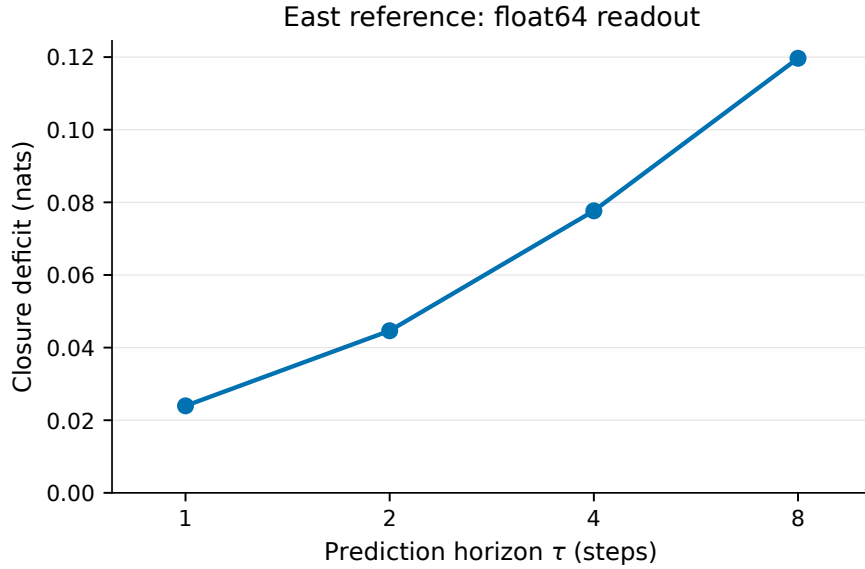


Figure 7. East-reference closure deficit (nats) across the declared τ ladder (run-only float64 readout; part of the frozen F2 reference). The frozen “deficit-decay” readout is the declared secant slope between two ladder points, not a fit; the deficit itself grows with horizon.

8.2 The scoreboard, unedited

[NEG.gt7_transfer_failures] Of the 44 committed predictions, 22 were scored against observed target cells: **8 passed and 14 failed**. The other 22 targeted cells that produced no observation. All 338 verdicts are filed unedited (Fig. 8). The failure pattern is itself a filed, tested fact:

- All 14 failures are FA-family interval predictions (5 t_K , 4 t_{peak} , 5 hump height), and every one is *one-sided high*: the observed FA value exceeds the predicted upper bound. FA humps are systematically later and larger than the East-anchored intervals allowed.
- Failures cluster at the deeper intermediate temperature: 12 of 14 failing cells have $\epsilon_{\text{mid}} = 2/5$, where every scored cell failed. At $\epsilon_{\text{mid}} = 1/2$, 7 of 9 scored interval cells passed (plus the passing order-relation prediction), and the two misses there are borderline (hump height 0.2033 against a bound of 0.2008; t_K of 5 against $[1, 3]$).
- All 22 unobserved cells are trap-family. The trap target’s exact hump search returned **found = false** in all 56 surface cells, with the uniform miss reason “does not start below equilibrium.” This is the dynamical corollary of the ϵ -invariance lemma of §3: the declared trap family’s equilibrium does not move with ϵ , so a temperature jump from equilibrium never leaves it below its new equilibrium, and the Kovacs protocol cannot produce a hump there at all.

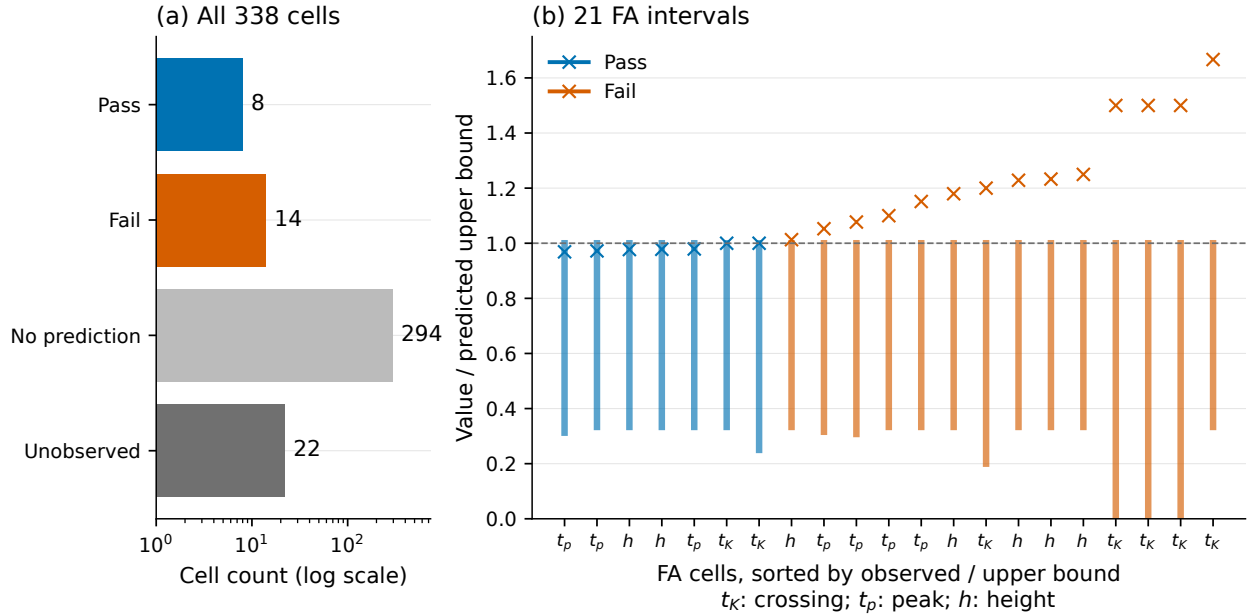


Figure 8. Left: verdict counts over all 338 cells (log scale). Right: the 21 scored FA interval cells (the 22nd scored FA prediction is the order relation, not plotted). Each interval and its observed value are divided by the interval’s upper bound, and cells are sorted by that ratio, so every failure lies above the dashed line at 1.

8.3 Reading the failures

The failures are first-class results: the protocol is designed to be falsifiable, and it was. Three readings are compatible with the data, and the protocol forbids choosing among them after the fact. (a) East-to-FA transfer of run-only constants degrades with quench depth, in the direction of FA responding slower and larger. (b) The declared rule for rescaling East intervals to FA, rather

than the readouts themselves, is what failed. (c) The trap column is not a partial failure but a structural mismatch between the protocol and the declared trap family, explained after the fact by the ϵ -invariance lemma. Under any reading, the readouts are what the run-only thesis says they are: readouts of runs, frozen, transferred and scored, not fitted. Score counts are not a fitted scaling law, and no post-hoc reinterpretation of failed intervals is permitted or performed.

9 GB9: The Lean-Checked Core and the Scope-Boundary Theorem

9.1 What is machine-checked

Lean [25] is a proof assistant: a theorem stated in it is accepted only when a machine has checked every step, and a development “compiles sorry-free” when no step has been left as an unproved placeholder. We use it for two things. The first is the abstract theorems the certificates rely on, for example that a witness pair exists if and only if the predictive description is strictly finer than the present one. The second is the logical consequences of the exact witness tables that the computation exports. Neither then depends on a reader trusting the prose. Precisely:

[GB9] Three Lean developments compile sorry-free on the pinned toolchain, `leanprover/lean4:4.29.0`: the vendored legacy `HolonomyMemory` core (paper [A]’s development [74]); the `DeclaredMemory` core, adapted from the upstream route-transport development by renaming modules and namespace only, with its mathematical declarations unchanged; and this project’s `GlassWitness` extension. All 96 theorems carry per-theorem axiom reports, regenerated from the live build, that match the stored audit. No theorem depends on anything beyond the standard axioms `propext`, `Classical.choice` and `Quot.sound`, and no `native_decide`, `sorry` or custom axiom appears. Per the framework’s fidelity legend, each row of the 97-row fidelity table (Appendix A; 96 theorems plus one row for evidence that lives outside Lean) carries its true label: 70 *verified*, 25 *narrowed/parametric* (the finite rational lookup witnesses and their generic constructors), 1 *schema projection*, and 1 *computational evidence only*. No schema or trivial declaration is described as a verified proof, and closure-deficit computations are not Lean theorems.

Among the verified rows are the witness/strict-refinement equivalence (`strictRefinement_iff_nonempty_predictiveWitness`) and the factorization theorems for future-sufficient states. Their direction matters: the minimal predictive quotient M factors, uniquely on the reachable image, through *any* future-sufficient state S . The converse need not hold, since S may carry redundant information.

9.2 The witness tables

The numeric witnesses are checked in the `DeclaredMemory` core as finite tables of exact rationals, exported by a generator from the Python computation:

- **NumericN10**: the GT2 Kovacs pair, current mean energy 20/7 and the two exact five-step future values.
- **NumericN8**: a mean-energy pair of the same kind on the $N = 8$ GT6 shell configuration (current mean energy 16/7).
- **EnergyN8**: the GT6 pair, with nine current events (the complete energy distribution, shared by both histories) and 21 future horizons.

For each, Lean checks the rational inequalities by kernel reduction and proves strict refinement, nonfactorization of the future through the current readout, and failure of future sufficiency for the

current readout. For **EnergyN8** this covers every horizon and every descriptor computed from the current energy law. Generic constructors prove the same consequences for any table with an actual rational split, and a constant-future negative control shows that a table without a split yields no strictness.

[NEG.lean_instance_structural] The boundary of this mechanization is precise. Lean checks the exported tables and what follows from them. It does not compute the East kernel, the preparations, the randomized-stop tuning, the null-vector selector, or the equality of the two current energy laws in GT6; those are verified in exact Python before export, and the fidelity table carries them as its one computational-evidence row. The legacy `glassInstance` remains a typed, index-based structural mirror.

9.3 The scope-boundary theorem

Why a second core? Because in the legacy core no witness can exist at all, and Lean proves it.

[FINDING.lean_collapse] For *every* instance of the legacy `RouteTransportCore`, the `observe_push` coherence field, combined with `CurrentEventEquiv`’s universal quantification over events, forces current equivalence to imply future-predictive equivalence (`current_implies_future_GENERIC`). Consequently the map from predictive to current classes is injective, and `PredictiveWitness`, `StrictRefinement` and `LoopAsymmetry` instances cannot be constructed in that core (`predictiveWitness_impossible_GENERIC`, `strictRefinement_impossible_GENERIC`, `predictiveToCurrent_injective_GENERIC`, `loopAsymmetry_impossible_GENERIC`; machine-checked, sorry-free).

The mechanism is simple. The legacy core’s naturality axiom lets every future event be pulled back to a present event, so “agreement on all present events” silently includes agreement on all pulled-back futures, and nothing is left for a witness to separate. The `DeclaredMemory` core drops the pullback and its coherence field: future observations are declared events of the pushed history rather than present events in disguise. That single change is what admits the tables above. The Python certificates never had the problem, because they quantify over a finite *declared* catalog of futures rather than over all pullback-closed events.

The collapse has a familiar shape. It is the proof pattern of strong lumpability for Markov chains [38, 15], where a local one-step coherence condition propagates an equivalence to all future times. It also has the form of biextensional collapse in the Chu-space sense [5, 56, 52], with `pullEvent` playing a role analogous to the Chu transform’s naturality condition; we offer this reading as an analogy, and no formal correspondence is claimed or proved. The instructive contrast is the causal-state construction of computational mechanics [63], where agreement on the present generally does *not* imply agreement on the future: the collapse is forced by the legacy core’s axiom, not by prediction itself.

10 GB10: Empirical Bridge Records

Nothing in the preceding sections is a measurement. The connection to the laboratory is made through three *bridge records*, drawn from the memory-effect phenomenology reviewed in [37], each of which states what a real instrument sees, what it suppresses, and under what threshold a measured signature would count as the laboratory counterpart of one of our certificates. Precisely:

[GB10] Three empirical bridge records connect the formal complex to laboratory phenomena, each filed with the complete five-field schema of F-II `prop:empirical-bridge` and validated by the bridge checker: (i) **Kovacs PVAc volume recovery** [40, 67, 65, 7], dilatometric specific-volume measurement under the temperature-jump/age/second-jump protocol, with the level map

tying the measured hump to the GT2 witness artifact by content hash; (ii) **spin-glass memory and rejuvenation** [36, 13, 79]; (iii) **colloidal-glass aging** [80, 22, 21, 53]. Each record carries: substrate and observable; instrument with a per-instrument visibility record (what the apparatus sees and what it suppresses; dilatometry, for instance, sees one-time specific volume and hump shape while suppressing the chain-conformation microstate); a level map with formal-artifact hashes; a threshold-witness relation stating when a measured signature *counts* as a laboratory-grade witness; and suppressed-structure nonclaims. Several numeric thresholds and source-precision fields (noise bands, drift thresholds, figure-level data locations) are explicitly marked **pending_source**: they await verification against the primary sources and are not yet usable thresholds.

The bridges are the paper’s only channel to the laboratory, and they are deliberately one-way: once its threshold is completed, a measured Kovacs signature that meets it is a laboratory-grade witness via the bridge, and nothing more is inferred from it. The laboratory counterpart of GT5 inherits GT5’s scope (§5): the most a measured Kovacs signature could license through the bridge is that the present description is incomplete, not that a single fictive-temperature-like scalar is insufficient. A laboratory replication targeting these records would need the declared protocol grid as an instrument-realizable schedule, a declared one-time observable panel, and a matched-pair preparation (two histories tuned to equal present panels). The GT6 construction shows one way to build such a pair: two randomized stops of a single aging run, mixed so that the present panel agrees exactly. A measured divergence of futures would then file as a laboratory-grade split pair through record (i)’s threshold-witness relation.

Nonclaim: No empirical-realization claim anywhere: laboratory data is never presented as a realization of the formal description. Visibility records are per instrument, with no cross-instrument merge into an instrument-free statement (F-II has no cross-instrument comparison theorem, and none is used).

11 Discussion

11.1 What is established, at which grade

Theorem-grade, each on its exactly declared finite class: the energy readout is not closed, with a margin over the nonzero equilibrium baseline certified in exact rational arithmetic (GT1); the Kovacs split-pair witness (GT2); foreclosure of all 512 energy-definable predicates, with a finer present-state readout as the exhibited separator (GT3); the kill-test panel, with five look-alike regimes rejected (GT4); $\text{MaxFiber} = 2$ with a saturating buyback ledger (GT5); the limiting-pressure bounds and the zero conditional limit on the GT6 shell; and, from the formalization, the collapse theorem for the legacy core (machine-checked). *Certificate-grade, shell-local*: GT6’s nonfactorization of the predictive description through the complete current energy law, over a 21-horizon window. *Run-only*: the GT7 protocol and its 8/14/294/22 scoreboard. *Recognition-grade*: the GB1 and GB2 demonstrations, the Lean fidelity filing, and the three empirical bridges. Nothing in this paper is graded higher than its artifact.

Taken together, the witnesses form a ladder. GT2 matches the mean energy of a Kovacs crossing against equilibrium. GT3 matches the complete energy law on a constructed reflection pair, separated by a present structural readout under the identity continuation. GT6 matches the complete energy law for two preparations of one aging run and lets the East dynamics separate them over 21 horizons. Each rung removes one way of saying “you just did not look at the energy carefully enough.” In every case the separating information is the arrangement of up spins, which the energy does not record; in GT2 and GT6 it is the East constraint that turns that arrangement into a different future.

11.2 What is open

Each open item has a register entry stating what closing it would take (`paper/gap_register.md`). None is load-bearing for a claim above, because every claim’s wording already excludes it.

[NEG.gb5_repair_sweep_not_built] The largest is GB5: no exhaustive sweep over a frozen class of candidate repairs exists, so no repair-impossibility certificate of any grade is claimed. “No admissible repair exists” is never asserted, and GT5’s buyback saturation is explicitly not a substitute.

The others:

- **GT6’s full filing.** The fixed-point stability gate fails on this shell because the packaging kernel has a unique stationary law. Restoring the full filing would need a different completion or limit construction, or a declared theory of approximate finite-time strata with its own tolerances.
- **A positive conditional pressure gap.** Proved impossible in the infinite-time limit for initial-law conditioning in this finite primitive model; the finite-time and killed-path contrasts are different objects.
- **Full-loop lawfulness.** The knockout panel is a diagnostic survey; a causal six-primitive ablation needs a common substrate and genuine mechanisms.
- **The GT2 loop certificate,** unfilled because the real loop’s history family does not close within six rounds.
- **GT1’s [B]-parity control** (skipped, filed) and **GT1’s breadth:** one declared (model, N , lens) cell, which is the declared-class discipline working as designed, but honestly a single cell.
- **GT7:** the interpretation of the failures (three readings left open by design), the absence of per-readout witnesses behind “run-only”, and the absence of retained evidence for the chronological order of the freeze stages.
- **GB1’s transient regime:** the corrected budget is stationary only.
- **Material forcing,** absent on this shell, so [C]’s fuller route is unexercised.
- **Bridge thresholds** marked `pending_source`.

Natural next experiments follow directly: a second registered prediction round with re-derived intervals and dated freezes; an Arrhenius-form trap family (per-trap rather than common temperature dependence) so that the Kovacs protocol is non-degenerate there; the FA, L_{panel} and $N=10$ extensions of GT1; a full-energy-law witness at larger N or with larger gaps; and laboratory matched pairs built by randomized stopping.

One ceiling is declared rather than open, per the framework’s recognition-open caveat: if glass aging carries a finite conservation or balance law over the declared records that supplies none of the six witness types, the six-role decomposition underlying this paper’s filings is conditional on the recognition hypothesis that no such feature exists. No such candidate was identified on the audited class; any future one files at recognition grade, separately.

11.3 Anti-reductionism, stated plainly

This paper’s certificates live at a boundary between two levels of description, the present-observable level and the predictive level, and the discipline that produced them cuts both ways. A passing certificate suite never proves emergence, and none of the information-theoretic emergence criteria in the literature [34, 60] is invoked. GT3 and GT6 say that the declared energy description provably cannot express the distinction that memory carries; that is the method working as intended, not a metaphysical discovery about glass, and not a reduction of either level to the other. Symmetrically, nothing here licenses reading the strictness of the memory level as autonomy, drive or an arrow of time: no P6_drive audit was performed, and a positive closure deficit is neither novelty nor irreversibility on its own. The paper makes no claim about the existence or non-existence of a thermodynamic ideal-glass transition, takes no side in the field’s thermodynamic-versus-dynamical debate [11], and makes no continuum-limit claims. Every quantifier in every claim ranges over a declared finite class, and the empirical legs are bridge claims only. The complete nonclaims ledger, every nonclaim of every claim row, unabridged, is printed in Appendix B.

12 Conclusion

On the exact-finite class built here, the observable one usually monitors for a glass does not form a closed description, and that failure is carried by certificates rather than intuitions. The closure deficit exceeds its nonzero equilibrium baseline by a margin proved in exact arithmetic. An exact Kovacs pair has the same present mean energy and different futures. Every predicate the energy lens can define agrees on a pair that one structural coordinate separates. Two randomized stops of one aging run share their entire energy distribution and still separate at every horizon of a 21-step window, so no descriptor of the energy law, fictive-temperature-like or otherwise, predicts the future there. A kill-test panel rejects its own constructed look-alikes, one added coordinate repairs each exhibited pair, and a registered transfer scoreboard is filed with its failures intact. Just as important are the results that do not hold: the fixed-point route to a fuller strict-extension filing fails on this shell, and an infinite-time conditional pressure gap is provably zero. Around the results sits the part we consider most transferable: a claims ledger tying every claim to hash-pinned evidence, the bindings machine-checked at build time, the wording human-audited and independently reviewed, and a nonclaims ledger printed in full.

The field kept asking “what is the order parameter of glass?” On the class where we can afford certificates, the answer is that the question is asked at the wrong level of description: the declared energy readout, down to its full distribution, cannot express the distinction that the witness pairs’ futures carry. The Kovacs effect has been such a witness pair, sitting in the literature since 1963. Whether that answer survives at larger sizes, richer observable panels and, ultimately, in the laboratory is what the gap register and the bridge records are built to test next.

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A Lean Fidelity Table

Every row of the fidelity table, generated from the live Lean build and its per-theorem axiom audit (§9). The module column gives the Lean file; “(exact Python)” marks the one row whose evidence lives outside Lean. Labels follow the framework’s five-value fidelity legend.

Table 7. Lean fidelity table, every row. Rows with no Lean encoding are labeled by the claim they document instead of a Lean identifier

module	Lean name / documented claim	fidelity
DeclaredMemory/ Asymmetry	predictiveToCurrent_eq_of_currentLoopTrivial	verified
DeclaredMemory/ Asymmetry	loopAsymmetry_exhibits_movedPredictive_ fixedCurrent	verified
DeclaredMemory/ Asymmetry	loopAsymmetry_exhibits_movedPredictive_ currentFixed	verified
DeclaredMemory/ Equivalences	currentEventEquiv_refl	verified
DeclaredMemory/ Equivalences	currentEventEquiv_symm	verified
DeclaredMemory/ Equivalences	currentEventEquiv_trans	verified
DeclaredMemory/ Equivalences	futurePredictiveEquiv_refl	verified
DeclaredMemory/ Equivalences	futurePredictiveEquiv_symm	verified
DeclaredMemory/ Equivalences	futurePredictiveEquiv_trans	verified
DeclaredMemory/ Equivalences	futurePredictiveEquiv_implies_currentEventEquiv	verified
DeclaredMemory/ Equivalences	currentEventEquiv_push	verified
DeclaredMemory/ Equivalences	currentEventEquiv_implies_future_of_all_ compatible	verified
DeclaredMemory/ Examples	bitCore_witness	verified
DeclaredMemory/ Examples	bitCore_swap_compatible	verified
DeclaredMemory/ Examples	bitCore_loop_asymmetry	verified
DeclaredMemory/ Examples	bitCore_read_not_compatible	verified
DeclaredMemory/ Loops	predictiveLoopAction_mk	verified
DeclaredMemory/ Loops	currentLoopAction_mk	verified
DeclaredMemory/ Loops	predictiveToCurrent_commutates_with_loopAction	verified
DeclaredMemory/ Sufficiency	predictiveQuotient_futureSufficient	verified
DeclaredMemory/ Sufficiency	futureSufficient_stateEq_implies_ futurePredictiveEquiv	verified

module	Lean name / documented claim	fidelity
DeclaredMemory/ Sufficiency	reachableStateRepresentative_state_eq	verified
DeclaredMemory/ Sufficiency	reachableStateToPredictiveQuotient_commutates	verified
DeclaredMemory/ Sufficiency	futureSufficient_factorsThroughPredictiveQuotient_onReachable	verified
DeclaredMemory/ Sufficiency	futureSufficient_factorization_unique_onReachable	verified
DeclaredMemory/ Sufficiency	reachableStateToPredictiveQuotient_unique	verified
DeclaredMemory/ Transport	futurePredictiveEquiv_push	verified
DeclaredMemory/ Transport	predictiveTransport_mk	verified
DeclaredMemory/ Transport	predictiveToCurrent_mk	verified
DeclaredMemory/ Transport	currentTransport_mk	verified
DeclaredMemory/ Transport	predictiveToCurrent_commutates_with_transport	verified
DeclaredMemory/ Witnesses	distinct_predictive_classes_same_current_class	verified
DeclaredMemory/ Witnesses	induces_strictRefinement	verified
DeclaredMemory/ Witnesses	strictRefinement_implies_predictiveWitness	verified
DeclaredMemory/ Witnesses	strictRefinement_iff_nonempty_predictiveWitness	verified
GlassWitness/ EnergyCore	strict_refinement	narrowed / parametric
GlassWitness/ EnergyCore	nonfactorization	narrowed / parametric
GlassWitness/ EnergyCore	current_law_not_futureSufficient	narrowed / parametric
GlassWitness/ EnergyCore	energy_descriptor_not_futureSufficient	narrowed / parametric
GlassWitness/ EnergyCore	constant_future_not_strict	narrowed / parametric
GlassWitness/ EnergyN8	all_horizons_split	narrowed / parametric
GlassWitness/ EnergyN8	strict_refinement	narrowed / parametric
GlassWitness/ EnergyN8	all_horizons_nonfactorization	narrowed / parametric
GlassWitness/ EnergyN8	current_law_not_futureSufficient	narrowed / parametric
GlassWitness/ EnergyN8	energy_descriptor_not_futureSufficient	narrowed / parametric
GlassWitness/Instance	glassPredictiveQuotient_futureSufficient	schema (projection)
GlassWitness/ NumericCore	strict_refinement	narrowed / parametric

module	Lean name / documented claim	fidelity
GlassWitness/ NumericCore	nonfactorization	narrowed / para- metric
GlassWitness/ NumericCore	probe_not_currentCompatible	narrowed / para- metric
GlassWitness/ NumericCore	current_panel_not_futureSufficient	narrowed / para- metric
GlassWitness/ NumericCore	index_state_futureSufficient	narrowed / para- metric
GlassWitness/ NumericCore	index_state_factors_onReachable	narrowed / para- metric
GlassWitness/ NumericCore	constant_future_not_strict	narrowed / para- metric
GlassWitness/ NumericN10	future_split	narrowed / para- metric
GlassWitness/ NumericN10	strict_refinement	narrowed / para- metric
GlassWitness/ NumericN10	nonfactorization	narrowed / para- metric
GlassWitness/ NumericN10	current_panel_not_futureSufficient	narrowed / para- metric
GlassWitness/ NumericN8	future_split	narrowed / para- metric
GlassWitness/ NumericN8	strict_refinement	narrowed / para- metric
GlassWitness/ NumericN8	nonfactorization	narrowed / para- metric
GlassWitness/ NumericN8	current_panel_not_futureSufficient	narrowed / para- metric
GlassWitness/ ScopeBoundary	current_implies_future_GENERIC	verified
GlassWitness/ ScopeBoundary	predictiveWitness_impossible_GENERIC	verified
GlassWitness/ ScopeBoundary	strictRefinement_impossible_GENERIC	verified
GlassWitness/ ScopeBoundary	predictiveToCurrent_injective_GENERIC	verified
GlassWitness/ ScopeBoundary	loopAsymmetry_impossible_GENERIC	verified
HolonomyMemory/ Asymmetry	predictiveToCurrent_eq_of_currentLoopTrivial	verified
HolonomyMemory/ Asymmetry	loopAsymmetry_exhibits_movedPredictive_ fixedCurrent	verified
HolonomyMemory/ Asymmetry	loopAsymmetry_exhibits_movedPredictive_ currentFixed	verified
HolonomyMemory/ Equivalences	currentEventEquiv_refl	verified
HolonomyMemory/ Equivalences	currentEventEquiv_symm	verified
HolonomyMemory/ Equivalences	currentEventEquiv_trans	verified
HolonomyMemory/ Equivalences	futurePredictiveEquiv_refl	verified

module	Lean name / documented claim	fidelity
HolonomyMemory/ Equivalences	futurePredictiveEquiv_symm	verified
HolonomyMemory/ Equivalences	futurePredictiveEquiv_trans	verified
HolonomyMemory/ Equivalences	futurePredictiveEquiv_implies_currentEventEquiv	verified
HolonomyMemory/ Equivalences	currentEventEquiv_push	verified
HolonomyMemory/ Loops	predictiveLoopAction_mk	verified
HolonomyMemory/ Loops	currentLoopAction_mk	verified
HolonomyMemory/ Loops	predictiveToCurrent_commutates_with_loopAction	verified
HolonomyMemory/ Sufficiency	predictiveQuotient_futureSufficient	verified
HolonomyMemory/ Sufficiency	futureSufficient_stateEq_implies_ futurePredictiveEquiv	verified
HolonomyMemory/ Sufficiency	reachableStateRepresentative_state_eq	verified
HolonomyMemory/ Sufficiency	reachableStateToPredictiveQuotient_commutates	verified
HolonomyMemory/ Sufficiency	futureSufficient_factorsThroughPredictiveQuotient_ onReachable	verified
HolonomyMemory/ Sufficiency	futureSufficient_factorization_unique_onReachable	verified
HolonomyMemory/ Sufficiency	reachableStateToPredictiveQuotient_unique	verified
HolonomyMemory/ Transport	futurePredictiveEquiv_push	verified
HolonomyMemory/ Transport	predictiveTransport_mk	verified
HolonomyMemory/ Transport	predictiveToCurrent_mk	verified
HolonomyMemory/ Transport	currentTransport_mk	verified
HolonomyMemory/ Transport	predictiveToCurrent_commutates_with_transport	verified
HolonomyMemory/ Witnesses	distinct_predictive_classes_same_current_class	verified
HolonomyMemory/ Witnesses	induces_strictRefinement	verified
HolonomyMemory/ Witnesses	strictRefinement_implies_predictiveWitness	verified
HolonomyMemory/ Witnesses	strictRefinement_iff_nonempty_predictiveWitness	verified
(exact Python)	Full East preparation, stopping-mixture construction and evolution	computational evi- dence only

B Full Nonclaims Ledger

Every nonclaim of every claims-ledger row, unabridged. Several are load-bearing: they fix how far a claim reaches, and the results sections cite them by row.

Table 8. Full nonclaims ledger: every nonclaim of every ledger row, unabridged

claim	nonclaim
GT1	GT1 certificate is scoped to East N=8, L_energy and the declared quench cell only; FA, L_panel and N=10 are undone extensions.
GT1	The tau ladder is the declared finite ladder [1, 2, 4, 8], not the full combinatorial grid.
GT1	Constrained East equilibrium under L_energy is a nonzero stationary CD baseline; only the unconstrained lumpable control is exact-zero.
GT1	The idempotence defect is a saturation diagnostic, not a closure criterion, and is distinct from the prototype-retention error.
GT1	[B] benchmark-parity control is filed as skipped in this artifact because the paper-specific example chains were not ported.
GT2	filed as P4 staged dependence, not a directionality or irreversibility claim
GT2	Only the mean energy is matched at the crossing; the full energy laws of the two histories differ.
GT2	Lean checks the exported rational table and its logical consequences, not the East kernel evolution that produced it.
GT2	Phase-0 Kovacs search output is a pre-certificate diagnostic, not a GT witness certificate.
GT2	The selected result freezes a later config but does not by itself prove a theorem-grade claim.
GT3	finite N=8 East foreclosure over Def(l_energy), not a continuum or all-lens claim
GT4	GT4 triage is over the declared finite benchmark family, not a shell-general theorem.
GT4	The fixtures are structural toys; they do not certify a thermal Kovacs coherence theorem or an exhaustive repair-class sweep.
GT4	Robustness fractions cover the three declared structural perturbations; they are not probabilistic confidence guarantees.
GT4	GB1 protocol-trap caveat remains a theorem-wrapper limitation; this table files the deterministic triage computation.
GT4	Phase 1 benchmark parity is a control suite, not a standalone GT4 triage certificate.
GT5	buyback saturation or nonsaturation is not a GB5 repair-impossibility certificate; GB5 requires a separate exhaustive repair-class sweep
GT5	the shipped Kovacs buyback curve is a minimal two-history witness with saturation_budget=1; it demonstrates buyback mechanics, not general single-scalar or TNM insufficiency
GT5	Loss is exact in-carrier sum squared error; no held-out generalization claim.
GT5	MaxFiber=2 does not lower-bound the number of real-valued coordinates needed to resolve memory in general.
GT6	GT6 is scoped to the audited glass shell, not to all East or glassy regimes.
GT6	No shell-general strict-bridge theorem is claimed.
GT6	The two sources are independent random stopping mixtures, an explicit carrier extension.
GT6	Finite-window separation at horizons 1..21 does not imply multiple exact packaging fixed points, all-time persistence, or a large-N statement.

claim	nonclaim
GT6	The tiny separating gap does not establish experimental detectability.
GT6	Strictness does not imply autonomous macro closure.
GT6	Strictness does not imply drive; P6_drive requires the separate GB1 audit discipline.
GT6	The pressure gap is not the strictness witness; the actual N8 energy-law split is.
GT6	This is a model realization, not a universal strict-extension theory.
GT6	Single bridge only; no stacked or glued promotions are claimed.
GT6	Spectral gaps and the legacy mixing-time field are window-calibration diagnostics, not proved total-variation bounds.
GT7	run-only readouts, not a fitted scaling law
GT7	'Run-only' is a procedural grade; no per-readout same-current, different-readout nonfactorization witness is claimed for the GT7 readouts.
GT7	Hash integrity establishes the frozen bytes, not chronological preregistration.
GT7	memdim_profile and deficit_decay transfer are out of scope; scored transfer covers hump readouts only.
GT7	Trap uses a declared depth observable, not East/FA spin-count energy.
GT7	transfer scoring is mechanical comparison against pre-registered predictions
GT7	score counts are not a fitted scaling law
GB1	GB1 is a recognition-grade demonstration and proof note, not a GT certificate.
GB1	The corrected budget is stationary only; no transient or arbitrary-initial-law bound is claimed.
GB1	The three-state demonstration in the artifact is float64, not exact.
GB1	The artifact payload records the general inequality as an open obligation; the counterexample certificate refutes its uncorrected stationary form and the proof note proves the corrected stationary budget.
GB2	GB2 demo records finite protocol-CD computations, not a GT claim certificate.
GB2	Windowed and marginalized CD values are float64 diagnostics; exactness is reserved for the lumpability check.
GB9	Full East kernel evolution, preparation and tuning are exact Python evidence, not Lean computations.
GB9	Lean's numeric strictness concerns the exported finite observation tables.
GB9	No shell-general, asymptotic or packaging-fixed-point theorem is mechanized.
GB9	The legacy glassInstance remains a structural index-based instance.
GB10	No empirical-realization claim anywhere; bridges are admissibility filings only.
GB10	Visibility records are per instrument; no cross-instrument merge into an instrument-free statement.
GB10	Fields marked pending_source are not verified thresholds.
FINDING.trap_epsilon_invariance	Not a novelty claim: the multiplicative-scalar cancellation is standard trap-model structure once stated; the contribution is stating and testing it for the declared family.
FINDING.trap_epsilon_invariance	Says nothing about trap dynamics or aging trajectories; only the stationary distribution is epsilon-invariant.
FINDING.trap_epsilon_invariance	Not a trap-literature no-go theorem: Arrhenius-form trap families can show Kovacs memory.
FINDING.lean_collapse	Not a claim that Kovacs memory is unformalizable; only that the legacy core's coherence axioms foreclose non-vacuous witness instances.
FINDING.lean_collapse	Does not weaken the exact-rational computational witnesses.
FINDING.gt6_pressure_limits	No positive infinite-time gap is claimed for initial-law conditioning.

claim	nonclaim
FINDING.gt6_ pressure_limits	The finite Jensen contrast is not a glass-specific strictness witness or a KL closure deficit.
FINDING.gt6_ pressure_limits	The killed survival contrast does not discharge a conditional-disintegration premise.
FINDING.gt6_ pressure_limits	Finite-ladder endpoints in the pressure-closure table are float diagnostics, not limiting pressures.
NEG.material_ forcing_absent	Absence of material forcing in this shell is not evidence about other shells or protocols.
NEG.gt6_original_ filing_failed	Finite predictive persistence is not used to mark the fixed-point stability gate as passed.
NEG.gt6_original_ filing_failed	No full-loop lawfulness or six-primitive causal activity is claimed.
NEG.gt7_transfer_ failures	Failure counts are not evidence against the run-only thesis; they are the run-only thesis operating as designed. No post-hoc reinterpretation of failed intervals is permitted.
NEG.gb5_repair_ sweep_not_built	'No admissible repair exists' is never asserted at any grade.
NEG.gt2_loop_ certificate_not_built	No loop-asymmetry or P3 route-mismatch claim is made anywhere on GT2's behalf.
NEG.gt1_b_parity_ control_open	No [B]-parity is claimed for GT1 until the control is actually run.
NEG.lean_instance_ structural	The Lean track certifies the abstract cores' theorems and the exported tables, not glass numerics end to end.

C Version History

v1 (23 September 2026). First release, DOI 10.5281/zenodo.22911309.

v2 (3 October 2026). This version, DOI 10.5281/zenodo.23118261. An independent mathematical and Lean audit of every certificate, with the following consequences.

- **GT6.** The strict-extension result is now the nonfactorization of the predictive description through the complete current energy law, on a new same-shell witness built from randomized stops (§7). The fuller filing's fixed-point stability gate is reported as failed, with a proof that the packaging kernel has a unique stationary law. The v1 verdict basis, which cited off-shell witnesses through verdict-level theorems, is withdrawn. The pressure diagnostic is replaced by certified limit bounds, a proof that the infinite-time conditional gap is zero, and separately labeled finite-time and killed-path contrasts.
- **GT1.** The margin is certified with exact probabilities and rational logarithm enclosures (v1: deterministic float64). The scope lists only the temperatures actually used, and the retention error is computed as its own quantity rather than copied from the idempotence defect.
- **Lean.** The numeric witnesses of GT2 and GT6 are mechanized as exact lookup tables in a declared-event core; v1 had no numeric witness in Lean. The fidelity table is generated from the live build (97 rows; v1: 17). The legacy collapse finding now includes explicit loop-asymmetry impossibility.

- **GB1.** The uncorrected stationary candidate bound is refuted by an exact counterexample and replaced by a proved budget with a contraction residual; the three-state demonstration is labeled float64 (v1 called it exact).
- **GT4.** The cleared protocol-trap and flattenable fixtures now preserve futures and support respectively, instead of erasing futures with a constant reset.
- **GT2, GT5, GT7, GB10.** The loop certificate's absence is explained by the real loop's non-closure; the GT5 loss is stated as a sum of squared errors; GT7 states what content hashes do and do not establish about chronology, and that scored transfer covers hump readouts only; bridge fields pending source verification are flagged.
- The exposition was rewritten throughout, with a new abstract, a results overview, a witness diagram and a figure for the GT6 witness. The GT7 scoreboard, its frozen registries and its scorer are unchanged.